

Section A

Answer **all** the questions in this section.

- 1 Stachyose is a storage oligosaccharide found in beans, peas and other legumes. Stachyose can only be metabolised by anaerobic microorganisms in the large intestine. It is responsible for the production of carbon dioxide, hydrogen sulfide and other gases, which collectively make up what is known as flatulence. The structure of stachyose is shown in Fig. 1.1

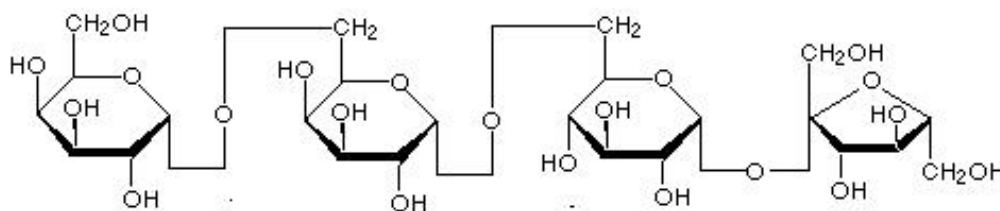


Fig. 1.1

- (a) Define the term glycosidic bond.

[1]

An enzyme known as α -galactosidase cleaves $\alpha(1,6)$ glycosidic bonds to sequentially release terminal galactose residues. The breakdown of stachyose by α -galactosidase is a two-step process. In the first step of this process, stachyose is hydrolysed to galactose and raffinose.

- (b) With reference to **Fig. 1.1**, draw the products of the first step in this hydrolysis. [2]

Microorganisms present in a rabbit's gut are able to digest carbohydrates in the plant material that they eat. Fig 1.2 shows the biochemical pathways by which cellulose and starch are digested in the gut of a rabbit.

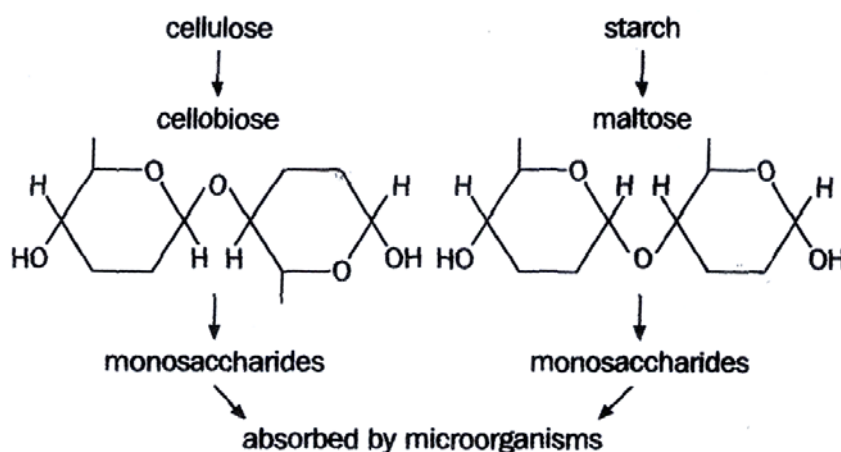


Fig 1.2

- (c) (i) Cellobiose and maltose are both disaccharides. [2]

With reference to Fig 1.2, explain why amylase enzymes produced by the rabbit are unable to digest cellobiose.

- (ii) Describe one way in which amylase enzymes lower the activation energy and increase the rate of hydrolysis of starch to maltose. [1]

- (d) Explain briefly how the structure of each polysaccharide is adapted for its role in a plant.

- (i) Starch [2]

- (ii) Cellulose [2]

[Total: 10]

2 Fig. 2.1 shows a few stages (A – F) of meiosis in the lily, *Lilium longiflorum*.

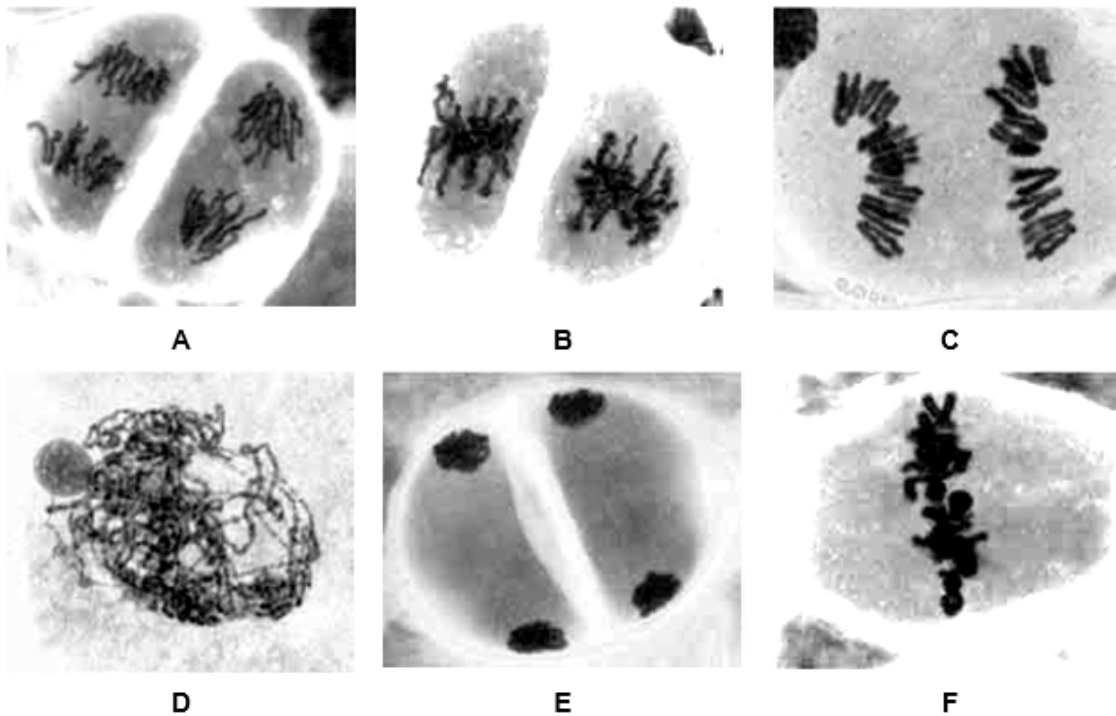


Fig 2.1

(a) With reference to Fig. 2.1,

- (i) arrange these stages in the correct sequence. [1]

- (ii) explain why stages **B** and **F** are different. [2]

- (b) Explain the importance of this nuclear division in sexual reproduction. [2]

- (c) Explain the significance of meiosis in genetic variation. [3]

Scientists suspected that colchicine, a drug that binds to tubulin, may arrest cells in meiosis. To test this hypothesis, cells of the *L. longiflorum* were treated with colchicine. The diameter and length of microtubules formed were analysed. The results of the experiment are shown in Fig. 2.2.

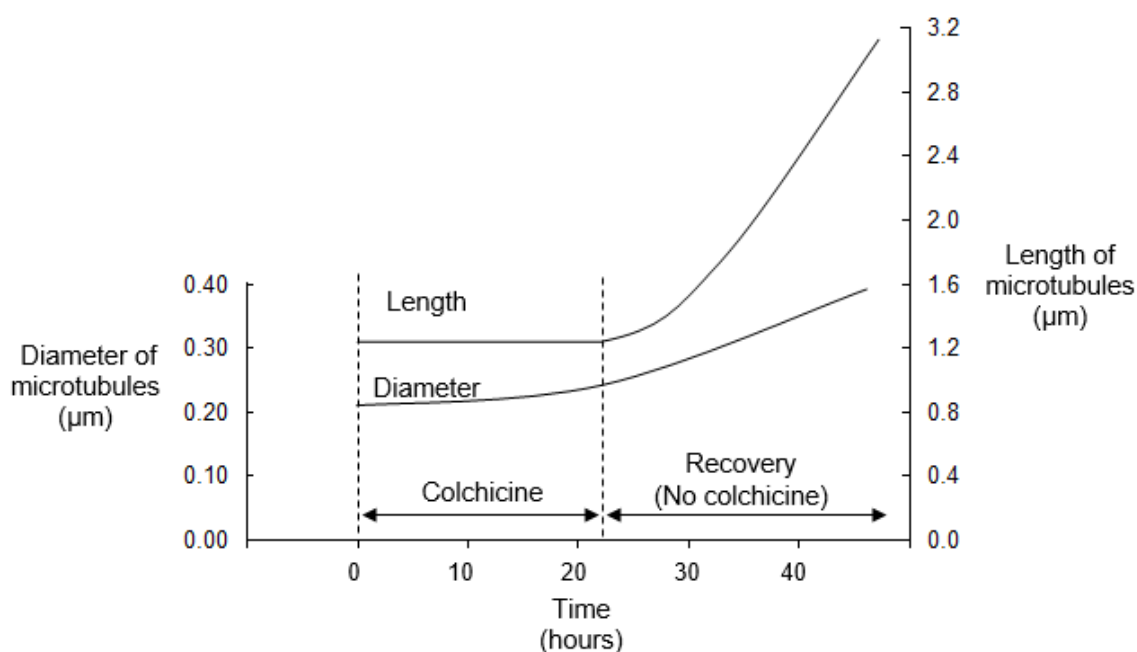


Fig 2.2

- (d) With reference to Fig. 2.2, describe and explain how colchicine is able to [2] inhibit meiosis.

[Total: 10]

- 3 Fig. 3.1 shows the growth pattern of *E. coli* cells in a medium containing a mixture of glucose and lactose. Observations of the concentrations of the two sugars in the medium show that glucose falls to very low levels (Fig. 3.1, circles) after the start of the experiment, but lactose remains relatively high and only starts to decrease near the end of the experiment (not shown in Fig. 3.1).

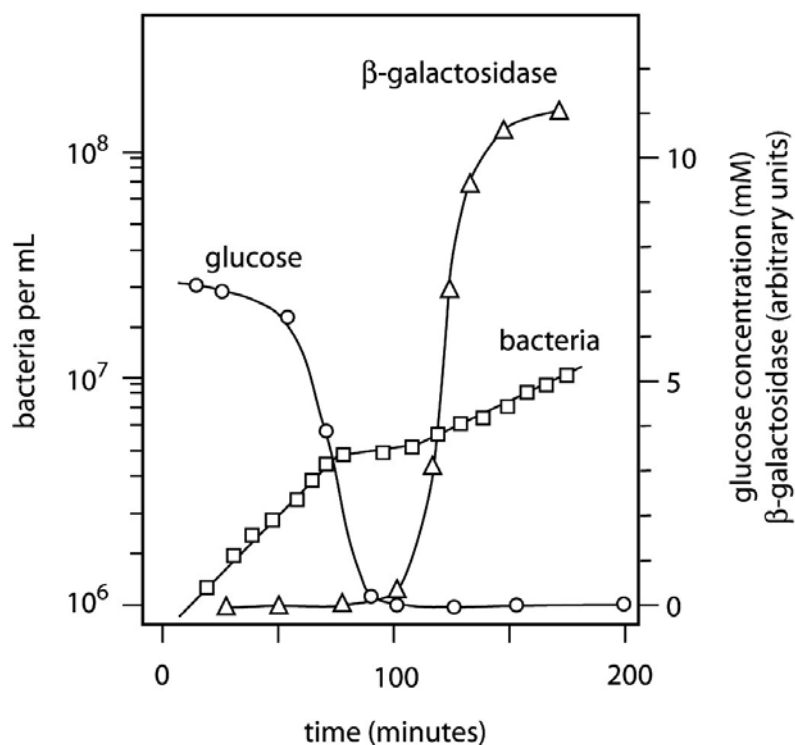


Fig. 3.1

- (a) With reference to Fig. 3.1, explain how expression levels of β -galactosidase is regulated by the lac repressor from

- (i) 25 minutes to 100 minutes;

[3]

- (ii) 100 minutes to 175 minutes. [3]

- (b) Describe another mechanism, which regulates the expression of β -galactosidase after 75 minutes. [2]

- (c) Explain why the number of bacteria cells remained constant between 75 minutes to approximately 110 minutes. [2]

- (d) β -galactosidase is coded for by a structural gene in the *lac* operon. [2]

Describe two differences between a structural gene and a regulatory gene in the *lac* operon.

[Total: 12]

- 4 Protein synthesis in a prokaryote is illustrated in Fig. 4.1.

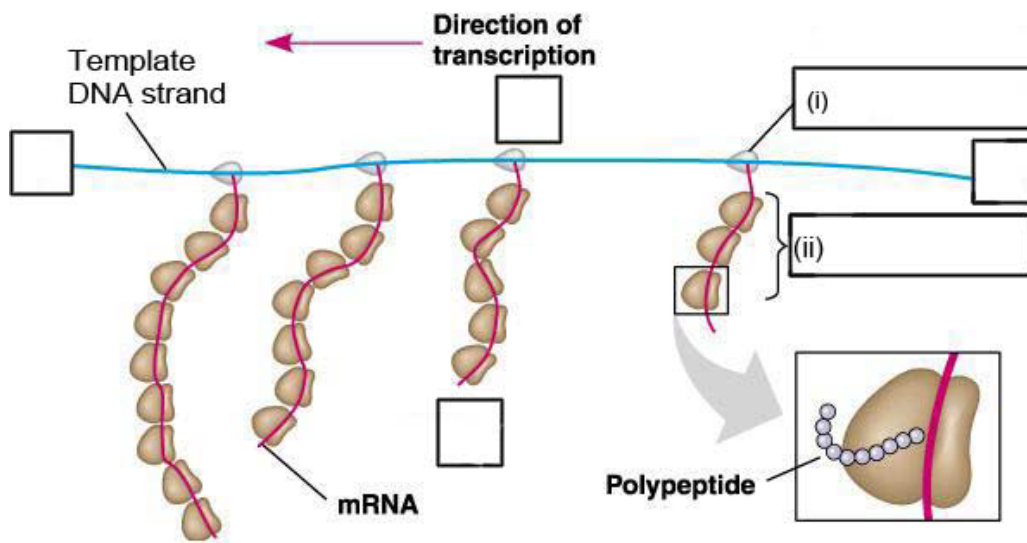


Fig. 4.1

- (a) Label the 5' and 3' ends of the mRNA and DNA template strand and [4]
structures (i) and (ii) on Figure 4.1.
- (b) State the main difference between the processes shown in Fig. 4.1 and that [1]
occurring in eukaryotes.
- (c) Explain your answers in (b). [2]

Over-expression of the HER2 (human epidermal growth factor receptor 2) gene is often associated with the development of breast cancer. One of the mechanisms causing the over-expression is illustrated in **Fig. 4.2**.

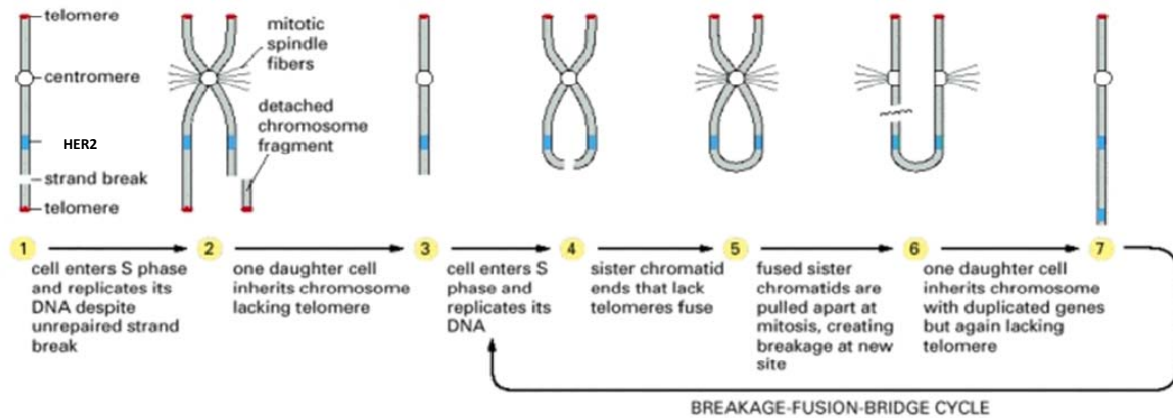


Fig. 4.2

(a) Name the process that has occurred.

[1]

(b) Explain how the above process can result in over-expression of HER2.

[3]

[Total: 11]

- 5 Research conducted with fruit flies, *Drosophila melanogaster*, has helped scientists elucidate many of the important concepts in genetics. The genome consists of four pairs of homologous chromosomes in females (males are the heterogametic sex).

On chromosome 2, there are three genes controlling three different phenotypic traits – eye colour, wing structure and body colour. Female *Drosophila* with brown eye (bn^+) and curly wings (cy^+) were mated to males with black body (bk^+). The F_1 were all wild-type. When the F_1 females were test crossed with males homozygous recessive for all three traits, the following result were obtained.

382	brown eyes, curly wings
401	black body
3	brown eyes
4	black body, curly wings
59	homozygous recessive for all three traits
67	wild-type
44	brown eyes, black body
40	curly wings

- (a) With reference to the information given above, explain why most of the [2]
offspring show parental phenotype.

Fig 5.1 shows the pair of homologous chromosomes 2 during prophase of meiosis of an F_1 female.

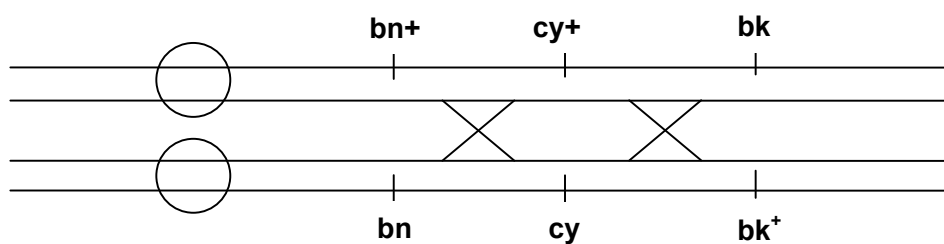


Fig 5.1

(b) With reference to Fig 5.1,

(i) calculate the recombinant frequency between genes cy^-/cy^+ and bk^-/bk^+ . [2]

(ii) calculate the recombinant frequency between cy^-/cy^+ and bn^-/bn^+ . [2]

(iii) State the two pieces of information required to genetic map of chromosome 2. [2]

A separate investigation on the size of the *D. melanogaster* was carried out. A cross was made between two *Drosophila* with a 0.5 cm body length and a 0.2 cm body length. Fig 5.2 shows the variation in *Drosophila* body length of the offspring from the cross.

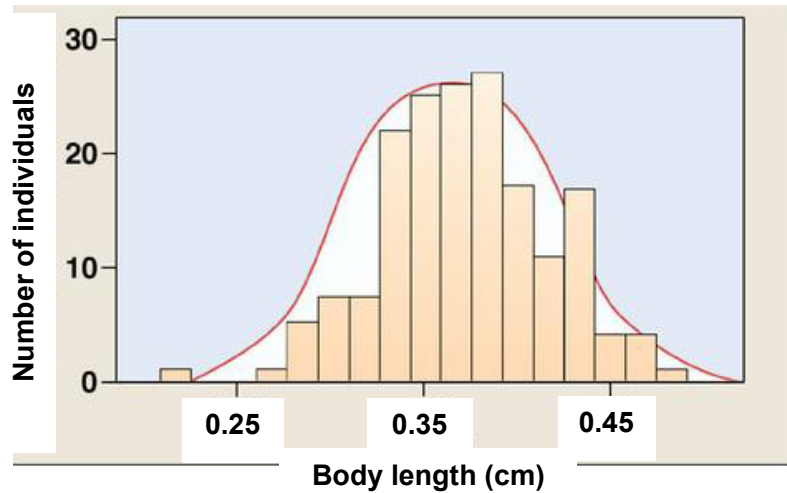


Fig 5.2

- (d) Explain why there is a range of phenotypes for this characteristic. [4]

[Total: 12]

6 Fig. 6.1 shows the cross section of an axon.

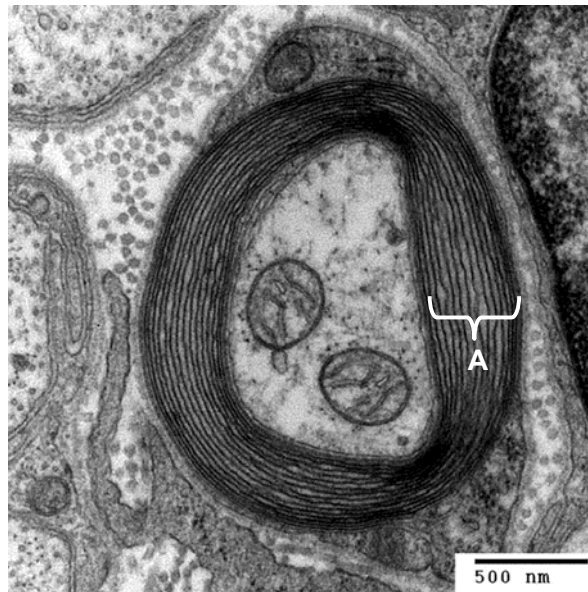


Fig. 6.1

(a) Explain how structure **A** is synthesised. [2]

The axon shown in Fig. 6.2 on page 17 is typical of patients suffering from multiple sclerosis. The membrane potential along a myelinated axon is measured using microelectrodes at points **P** to **U**, which corresponds to regions without myelin. The bottom of the graph is a schematic diagram showing the relative positions of points **P** to **U** along the axon.

(Note that microelectrodes at points **T** and **U** did not register a change in membrane potential and has been omitted from Fig. 6.2 for clarity purposes.)

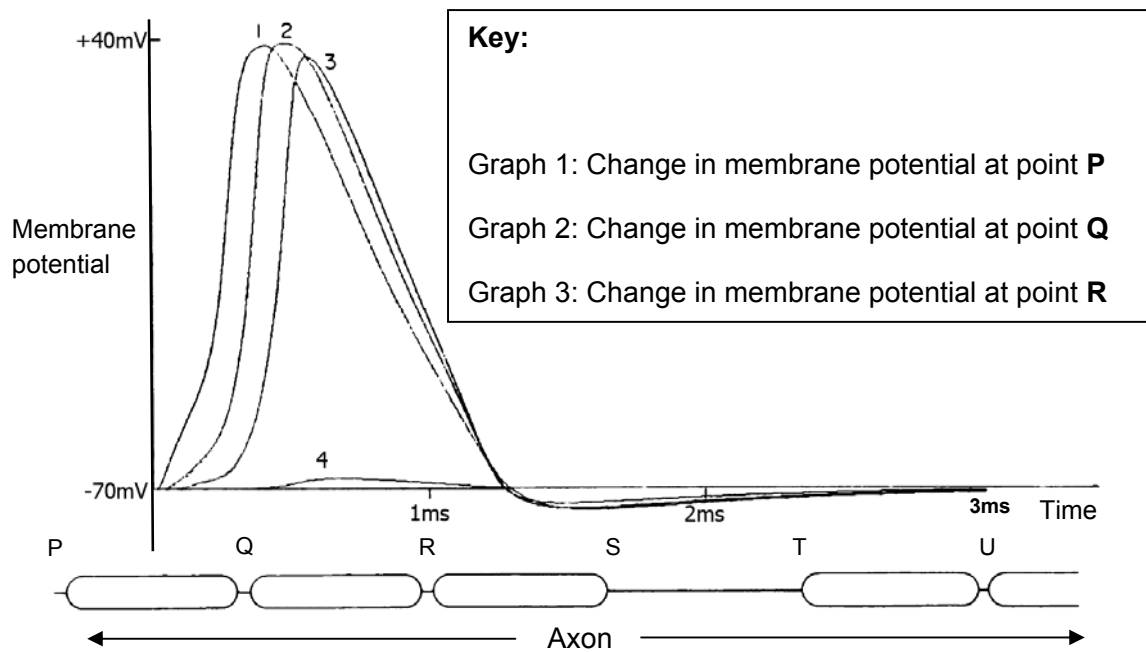


Fig. 6.2

(b) With reference to Fig. 6.2,

- (i) describe and explain the change in membrane potential at point **P** of the axon. [3]

- (ii) describe and explain the change in membrane potential at point **S**. [2]

The neurones in the multiple sclerosis patients are different in that the antibodies for myelin are present in high concentrations, causing an autoimmune response that leads to partial degeneration of the myelin sheath.

- (c) Explain why microelectrodes at points **T** and **U** did not register a change in membrane potential. [2]

Table 6.1 shows the diameter and transmission rate of action potential for different types of neurones.

Table 6.1

Type of neurone	Presence of myelin	Average diameter (µm)	Average transmission rate (m/s)
Sensory	No	2.12	3.9
Sensory	Yes	4.05	8.4
Motor	No	1.89	2.9
Motor	Yes	4.20	10.9
Relay	No	2.03	3.6
Relay	Yes	3.89	8.1

- (d) Using Table 6.1, explain how the presence of myelin and the diameter of the axon affect the transmission of a nerve impulse. [2]

In an anti-smoking campaign, the success of the campaign is measured by the number of people who pledged to quit smoking. Quitting smoking requires more than willpower. Investigators attempt to understand and demystify the physiological effects of nicotine.

The first step was to trace the movement of nicotine by introducing radioactively labelled nicotine into the brain and see which type of carrier protein it binds to. To the surprise of many, it binds to the receptor directly (a subset of receptors for acetylcholine, on Fig. 6.3) on a post-synaptic nerve cell.

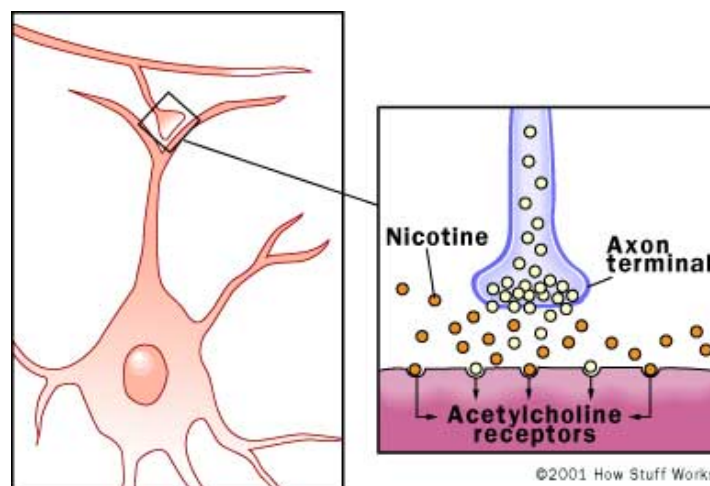


Fig. 6.3

- (e) Suggest why nicotine could bind to the acetylcholine receptors. [1]

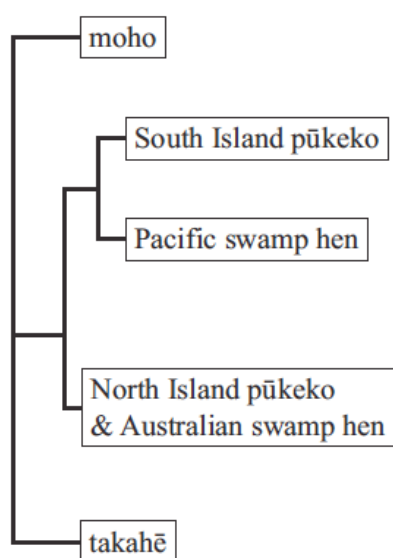
[Total: 12]

- 7 The pūkeko (*Porphyrio melanotus*) and takahē (*Porphyrio hochstetteri*) look similar and share a common ancestor, but have evolved differently. Their flighted ancestor came over from Australia millions of years ago.

Fossil evidence suggests the takahē have been present in New Zealand lowlands and forest margins for at least 5 million years, before the presence of ground-dwelling mammals and human arrival in New Zealand. It is flightless, a slow breeder and is now endangered. The takahē was thought to be extinct, but a small population with low genetic variation was discovered in 1948 among the remote Fiordland mountains of the South Island. There are an estimated 227 adult birds in Fiordland's isolated alpine habitat and on four predator-free offshore islands. The North Island takahē or moho is extinct.

The pūkeko invaded New Zealand within the past 1,000 years. A cooperative breeder*, this flighted bird has become well established within open grassland, swamps, and agricultural lands.

Recent DNA evidence points to the pūkeko in the North Island having close connections with the Australian swamp hen, whereas South Island pūkeko show close relation to the Pacific swamp hen. The lineage is shown in Fig. 7.1.



The takahē



The pūkeko

Fig. 7.1

***Cooperative breeding** is a social system where helpers provide care for offspring produced by breeding group members. For the breeding pair, their primary benefit is an increased reproductive rate and survival.

- (a) Complete Table 7.2 to show the classification of the pūkeko. [2]

Table 7.2

Kingdom	
	Chordata
	Aves
Order	Gruiformes
	Rallidae
Genus	
	<i>Porphyrio melanotus</i>

- (b) Differences in the DNA sequences that encode cytochrome b in the takahē and the moho from New Zealand have been measured, using museum specimens and samples taken from the Fiordland population.

- (i) When differences in the cytochrome b gene are plotted against time [2]
since colonisation, the plot of the line is straight. Describe how this
supports the neutral theory of molecular evolution.

- (ii) Suggest why the cytochrome b gene is used to measure changes in [2]
DNA sequences between the two species of flightless birds.

- (c) A biologist wanted to establish the order in which each species had evolved from the common ancestor. DNA hybridisation between the various species was carried out. After analysing the results, the scientist arrived at the phylogenetic tree shown in Fig. 7.1. [2]

Using your knowledge on DNA hybridisation, described the results obtained from the technique that would have led the biologist to establish the phylogeny between the North and South Island pūkeko and the Australian and Pacific swamp hens.

- (d) Using the information provided in the question and your knowledge on evolution, compare and contrast how evolutionary selection pressures have shaped the evolution and survival of the pūkeko and takahē differently. [5]

[Total: 13]

Section B

Answer **one** question.

Write your answers on the separate answer paper provided.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where applicable.

Your answers must be set out in sections **(a)**, **(b)** etc., as indicated in the question.

- 8** **(a)** Explain the implication of mutation with respect to phenotype in cystic fibrosis. [7]
- (b)** Outline how the rate of protein synthesis in eukaryotes may be controlled. [7]
- (c)** Describe the roles of the different RNA molecules involved in translation. [6]
- 9** **(a)** Explain the implication of checkpoint dysregulation with respect to cell division in mitosis. [7]
- (b)** Outline the structural features of two named nucleic acids involved in protein synthesis. [7]
- (c)** Describe how a normal cell can form a malignant tumour. [6]